

Recursive Estimation	
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1 Probability Theory	
1.1 Random Variables	
1.1.1 Discrete Random Variables	
DRV – Discrete Random Variable defined by $\mathcal{X}, p_x: \mathcal{X} \rightarrow \mathbb{R}$	
$\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x} \in \mathcal{X}} p_{x y}(\bar{x} \bar{y}) = 1$	
Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy}(\bar{x}, \bar{y})$
Cond. / Product Rule	$p_{x y}(\bar{x} \bar{y}) := \frac{p_{xy}(\bar{x}, \bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y} \in \mathcal{Y}} p_{x y}(\bar{x} \bar{y})p_y(\bar{y})$
1.1.2 Continuous Random Variables	
CRV – Continuous Random Variable: $\text{DRV} \sum \rightsquigarrow \text{CRV} \int$	
- Set $\mathcal{X} \subset \mathbb{R}$ section of real line - PDF p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$	
Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a, b)) = \int_a^b p(\bar{x})d\bar{x}$	
1.1.3 Conditional PDF	
Conditional PDF	
Margin. / Sum Rule	$p(\bar{x} \bar{z}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy z}(\bar{x}, \bar{y} \bar{z})$
Cond. / Product Rule	$p_{x yz}(\bar{x} \bar{y}, \bar{z}) := \frac{p_{xy z}(\bar{x}, \bar{y} \bar{z})}{p_{y z}(\bar{y} \bar{z})}$
Independence	
$p(x y) = p(x) \Leftrightarrow p(y x) = p(y) \Leftrightarrow p(x, y) = p(x)p(y)$ $p(x, y, z) = p(x, y z)p(z) \rightarrow x, y$ indep. $p(x, y z) = p(z x, y)$	
Conditional Independence	
The knowledge of z makes x and y independent:	
$p(x y, z) = p(x z) \Leftrightarrow p(y z) = p(x z)p(y z)$	
Caution!!! in general we still have: $p(x, y) \neq p(x)p(y)$	
Caution!!! Independence $\nRightarrow$ Conditional Independence	
1.2 Expectation and Variance	
1.2.1 Expectation	
Definition: <i>Integral for CRV</i>	
$\mathbb{E}_x[x] = \sum_{\bar{x} \in \mathcal{X}} \bar{x}p_x(\bar{x})$	
Linearity: $\mathbb{E}_{xy}[a + bx + cy] = a + b\mathbb{E}_x[x] + c\mathbb{E}_y[y]$ Multi Variable: $\mathbb{E}_{xy}[g(x, y)] = \sum_{\bar{y}} \sum_{\bar{x}} \bar{x}g(\bar{x}, \bar{y})p_{xy}(\bar{x}, \bar{y})$ Independence of x, y $\Rightarrow \mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\mathbb{E}_y[y]$	
Law of Unconscious Statistician for $y = g(x)$	
$\mathbb{E}_y[y] = \sum_{\bar{y} \in \mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x} \in \mathcal{X}} g(\bar{x})p_x(\bar{x})$	
1.2.2 Variance (generally a matrix)	
$\text{Var}_x[x] = \mathbb{E}_x[(x - \mathbb{E}_x[x])(x - \mathbb{E}_x[x])^\top] \stackrel{1D}{=} \mathbb{E}_x[x^2] - \mathbb{E}_x[x]^2$	
Linearity	
$\text{Var}_x[a + bX + cY] = b^2 \text{Var}_X[X] + c^2 \text{Var}_Y[Y] + 2bc \text{Cov}[X, Y]$	
Covariance: if independent $\text{Cov}(X, Y) = 0$	
$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	
1.3 Distribution Sampling	
CDF – Cumulative Distribution Function	
$\hat{F}_x(\bar{x}) := \sum_{i=-\infty}^{\bar{x}} \hat{p}_x(i) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\bar{x}} p_x(\hat{x})d\hat{x}$	
A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that:	
$\hat{F}_x(\bar{x} - 1) < \bar{u} \leq \hat{F}_x(\bar{x})$	
A2 – Multiple DRV or CRV	
Works for finite & infinite # of elements	

- Decompose $\hat{p}(\bar{x}, \bar{y}) = \hat{p}_{x y}(\bar{x} \bar{y})\hat{p}_y(\bar{y})$ - Apply A1 to get sample $\bar{y}$ via $\hat{p}_y(\bar{y})$ - With $\bar{y}$ fixed, apply A1 to get sample $\bar{x}$ via $\hat{p}_{x y}(\bar{x} \bar{y})$ A3 – 1 CRV $\hat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0, 1)$ and solve: $\bar{u} = F_x(\bar{x}) \rightarrow \bar{x} = F_x^{-1}(\bar{u})$	
1.4 Change of Variables	
DRV: $x = g(y)$	
$p_x(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}: g(\bar{y}) = \bar{x}} p_y(\bar{y})$	
CRV general case $y = g(x), \rightarrow x = g^{-1}(y)$:	
$p_y(\bar{y}) = \left \frac{dg}{dx}(\bar{x}) \right , \quad \bar{x} = g^{-1}(y)$	
CRV – Multivar. & Indep. Case: $z = x + y$	
$p_z(\bar{z}) = \sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$	
CRV – Multivar. & Cond. Case: $z = g(w, x), w = h(z, x)$	
Assumptions: - $g(w, x), p_w$ cont. diff'bar - $g(w, x)$ strictly monotonic - For scalars $\det(x)$ is just x	
$p_{z x}(\bar{z} \bar{x}) = p_w(h(\bar{z}, \bar{x}) \bar{x}) \cdot \left \det \left(\frac{\partial}{\partial w} g(h(\bar{z}, \bar{x}), \bar{x}) \right) \right ^{-1}$	
1.5 Miscellaneous	
Indep. multiplication: $z = xy \rightarrow p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$	
Uniform Distribution	
$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases} \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \frac{a+b}{2} \\ \text{Var}\{x\} &= \frac{(b-a)^2}{12} \end{aligned} \right.$	
Triangular Distribution	
$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$	
$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$	
$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \text{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$	
Gaussian Distribution $\mathbf{x} = (x_1, \dots, x_D) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	
$p_{\mathbf{x}}(\bar{\mathbf{x}}) = \frac{1}{(2\pi)^{D/2} \det(\boldsymbol{\Sigma})^{1/2}} \exp \left(-\frac{(\bar{\mathbf{x}} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})}{2} \right)$	
$\rightsquigarrow$ fully characterized by mean $\boldsymbol{\mu} \in \mathbb{R}^D$ and Variance $\boldsymbol{\Sigma} \in \mathbb{R}^{D \times D}$	
(Scalar: $D = 1, \sigma^2 = \det(\boldsymbol{\Sigma}) \rightarrow \det(\boldsymbol{\Sigma})^{1/2} = \sigma$)	
$p(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \mu \\ \text{Var}\{x\} &= \sigma^2 \end{aligned} \right.$	
Affine transformation of Gaussian: $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	
$\mathbf{y} = M\mathbf{x} + \mathbf{b} \rightarrow \mathbf{y} \sim \mathcal{N}(M\boldsymbol{\mu} + \mathbf{b}, M\boldsymbol{\Sigma}M^T)$	
Linear combination of jointly GRV: $\mathbf{x}, \mathbf{y}$	
$\mathbf{z} = M_x\mathbf{x} + M_y\mathbf{y} \Rightarrow \mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$	
$\boldsymbol{\mu}_z = M_x\boldsymbol{\mu}_x + M_y\boldsymbol{\mu}_y$	
$\boldsymbol{\Sigma}_z = M_x\boldsymbol{\Sigma}_xM_x^T + M_x\boldsymbol{\Sigma}_{xy}M_y + M_y\boldsymbol{\Sigma}_{xy}^TM_x + M_y\boldsymbol{\Sigma}_yM_y^T$	
(Red symbols are zero if $\mathbf{x}, \mathbf{y}$ are independent)	
Diagonal Variance $\rightsquigarrow y_i$ mut.indep. iff $\boldsymbol{\Sigma}$ diagonal	
$p(y) = \prod_{i=1}^D \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right)$	

2 Bayesian Tracking	
2.1 Bayes Theorem	
Definition: (Applies to discrete and continuous random variables)	
$p(x y) = \frac{p(y x)p(x)}{p(y)}$	
x =system state, $p(x)$ =prior belief of state, z =observations related to state $p(z x)$ =observation likelihood, $p(x z)$ =posterior belief of state $p(z)$ =normalization constant $= \sum_x p(z x)p(x)$ For multiple observations:	
$p(x z_1, \dots, z_N) = \frac{\overbrace{p(x)}^{\text{prior}} \prod_i \overbrace{p(z_i x)}^{\text{observation likelihood}}}{\underbrace{\sum_{x \in \mathcal{X}} p(x) \prod_i p(z_i x)}_{\text{normalization}}}$	
2.2 Bayesian Tracking	
Model	
Objective: Calculate pdf of state $p(x(k) z(1:k))$	
$x(k) = q_{k-1}(x(k-1), v(k-1)) \in \mathcal{X}$	
$z(k) = h_k(x(k), w(k))$	
Prior Update - State Prediction	
$p(x(k) z(1:k-1)) = \sum_{x(k-1)} \underbrace{p(x(k) x(k-1))}_{\text{process model}} \underbrace{p(x(k-1) z(1:k-1))}_{\text{previous iteration}}$	
Measurement Update	
$p(\bar{x}(k) \bar{z}(1:k)) = \frac{\overbrace{p(\bar{z}(k) \bar{x}(k))}^{\text{measurement model}} \overbrace{p(\bar{x}(k) \bar{z}(1:k-1))}^{\text{prior}}}{\underbrace{\sum_{i \in \mathcal{X}} p(\bar{z}(k) i)p(i \bar{z}(1:k-1))}_{\text{normalization}}}$	
Computer Implementation	
1. Enumerate state space $\mathcal{X} = \{0, 1, \dots, N-1\}$ 2. Posterior define $\mathbf{a}_{k k}^i := p_{x(k) z(1:k)}(i \bar{z}(1:k))$ 3. Prior define $\mathbf{a}_{k k-1}^i := p_{x(k) z(1:k-1)}(i \bar{z}(1:k-1))$	
Initialization $\mathbf{a}_{0 0} = p_{x(0)}(i) \quad i = 0, \dots, N-1$	
$\left. \begin{aligned} \mathbf{a}_{k k-1}^i &= \sum_{j=0}^{N-1} p_{x(k) x(k-1)}(i j)\mathbf{a}_{k-1 k-1}^j \\ \mathbf{a}_{k k}^i &= \frac{p_{z(k) x(k)}(\bar{z}(k) i)\mathbf{a}_{k k-1}^i}{\sum_{j=0}^{N-1} p_{z(k) x(k)}(\bar{z}(k) j)\mathbf{a}_{k k-1}^j} \end{aligned} \right\} \text{Iterate } k > 0$	
3 Extracting Estimates	
3.1 General Measurement	
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$	
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent	
3.2 General Measurement	
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m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent	
3.3 Likelihood Estimators	
3.3.1 ML-Maximum Likelihood	
Choose x to make $\bar{z}$ most likely:	
$\left. \begin{aligned} x \in \mathcal{X} & \text{ unknown} \\ p_x(\bar{x}) & \text{ unknown} \end{aligned} \right\} \hat{x}^{ML} := \underset{\bar{x} \in \mathcal{X}}{\operatorname{argmax}} [p_{z x}(\bar{z} \bar{x})]$	
$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF	
3.3.2 MAP-Maximum A Posteriori	

ML given observation & prior belief about x :	
$\left. \begin{aligned} x \in \mathcal{X} & \text{ unknown} \\ p_x(\bar{x}) & \text{ known} \end{aligned} \right\}$	$\hat{x}^{MAP} := \underset{\bar{x} \in \mathcal{X}}{\operatorname{argmax}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]$
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$	
3.4 General Measurement	
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$	
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent	
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$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF	
3.5.2 MAP-Maximum A Posteriori	
ML given observation & prior belief about x :	
$\left. \begin{aligned} x \in \mathcal{X} & \text{ unknown} \\ p_x(\bar{x}) & \text{ known} \end{aligned} \right\}$	$\hat{x}^{MAP} := \underset{\bar{x} \in \mathcal{X}}{\operatorname{argmax}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]$
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$	
3.6 Least Squares Estimators	
3.6.1 Linear Least Squares	
ML estimate when errors are indep. 0-mean, smae variance, $\mathcal{N}$ -distributed:	
$\hat{x}^{LS} := \underset{x}{\operatorname{argmin}} [(z - Hx)^\top (z - Hx)] = (H^\top H)^{-1} H^\top z$	
$\hat{x}^{LS} = \hat{x}^{ML}$ for gaussian noise	

3.6.2
Weighted Least Squares

$$\hat{x}^{WLS} := \operatorname{argmin}_x \left[(\bar{z} - H\hat{x})^\top R^{-1} (\bar{z} - H\hat{x}) \right]$$

$$= (H^\top R^{-1} H)^{-1} H^\top R^{-1} \bar{z}$$

Incorporate Prior Knowledge on x:
 $r \sim \mathcal{N}(0, P_x) \Rightarrow x := \hat{x} + r, \mathbb{E}\{x\} = \hat{x}_0, \operatorname{Var}\{x\} = P_x$

Introduce Extended System:

$$\tilde{z} = \begin{bmatrix} \hat{x}_0 \\ z(1:k) \end{bmatrix}, \tilde{H} = \begin{bmatrix} \mathbb{I} \\ H(1:k) \end{bmatrix} \tilde{w} = \begin{bmatrix} -r \\ w(1:k) \end{bmatrix}, \tilde{R} = \begin{bmatrix} P_x & 0 \\ 0 & R(1:k) \end{bmatrix}$$

3.6.3 Recursive Least Squares

Interpretation Update $\hat{x}$ recursively when new $\bar{z}(k)$ arrives
Estimation Error: $e(k) = (\mathbb{I} - KH) e(k-1) - Kw(k)$
Objective Minimize MSE with $P(k) = \operatorname{Var}\{e(k)\}$

$$J(k) := \mathbb{E} \left\{ e^\top(k) e(k) \right\} = \mathbb{E} \{ \operatorname{trace}(P(k)) \}$$

Algorithm

Initialization: $\hat{x}(0) = \hat{x}_0, \quad P(0) = P_x = \operatorname{Var}\{x\}$

Recursion: Observe $\bar{z}(k)$, then update:

$$K(k) = P(k-1)H(k)^\top \left(H(k)P(k-1)H(k)^\top + R(k) \right)^{-1}$$

$$\hat{x}(k) = \hat{x}(k-1) + K(k) (\bar{z}(k) - H(k)\hat{x}(k-1))$$

$$P(k) = (\mathbb{I} - K(k)H(k))P(k-1)(\mathbb{I} - K(k)H(k))^\top + K(k)R(k)K(k)^\top$$

4
Kalman Filter

4.1 Kalman Filter

Exact Solution for Bayes Tracking of LTI Systs. with Gauss. Noise.

Model (Linear Time-Invariant)

$$x(k) = A(k-1)x(k-1) + u(k-1) + v(k-1)$$

$$z(k) = H(k)x(k) + w(k)$$

$x(0), v(\cdot), w(\cdot)$ ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

Auxilliary Variables

Problem Formulation want $p_{x(k)|z(1:k)} \rightsquigarrow$ bayesian tracking provides this **BUT** we have **CRVs** and don't want integrals
 $\Rightarrow$ exploit linearity, GRV to convert to matrix manipulations
Auxilliary Variables:

Init: $x_m(0) = x(0)$

$$\mathbf{S1:} \; x_p(k) = A(k-1)x_m(k-1) + u(k-1) + v(k-1)$$

$$\mathbf{S2:} \; z_m(k) = H(k)x_p(k) + w(k)$$

$$p_{x_m(k)}(\xi) := p_{x_p(k)|z_m(k)}(\xi|\bar{z}(k)) \quad \forall \xi$$

Equivalency x conditioned on $z(1:k-1)/z(1:k)$

$$p_{x_p(k)}(\xi) = p_{x(k)|z(1:k-1)}(\xi|\bar{z}(1:k-1))$$

$$p_{x_m(k)}(\xi) = p_{x(k)|z(1:k)}(\xi|\bar{z}(1:k))$$

GRV x_p and x_m are GRVs for $k = 1, 2, \dots$

$$\hat{x}_p(k) := \mathbb{E}\{x_p(k)\} \qquad P_p(k) := \operatorname{Var}\{x_p(k)\}$$

$$\hat{x}_m(k) := \mathbb{E}\{x_m(k)\} \qquad P_m(k) := \operatorname{Var}\{x_m(k)\}$$

Recursive Update Equations

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + Q(k-1)$$

S2 – A Posteriori Update / Measurement Update

$$P_m(k) = \left(P_p^{-1}(k) + H^\top(k)R^{-1}(k)H(k) \right)^{-1}$$

$$\hat{x}_m(k) = \hat{x}_p(k) + \underbrace{P_m(k)H^\top(k)R^{-1}(k)}_{\text{correction factor}} \underbrace{(\bar{z}(k) - H(k)\hat{x}_p(k))}_{\text{prediction error}}$$

Alternative Equations – all matrices/vectors at time k

S2 – Kalman Filter Sqiare:

$$K = P_p H^\top \left(H P_p H^\top + R \right)^{-1}$$

$$\hat{x}_m = \hat{x}_p + K (\bar{z} - H\hat{x}_p)$$

$$P_m = \underbrace{(\mathbb{I} - KH) P_p}_{\text{cheaper}} = \underbrace{(\mathbb{I} - KH) P_p (\mathbb{I} - KH)^\top + K R K^\top}_{\text{fewer numerical errors}}$$

Equivalent to RLS $\rightsquigarrow$ among linear, unbiased estimators KF minimizes MSE

KF Remarks

Known Data $A(k), H(k), Q(k), R(k) \; \forall k$ as well as $P_0 \rightsquigarrow$ KF matrices $P_p(k), P_m(k), K(k)$ can be computed offline

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$

$$\hat{x}_m(k) = \underbrace{(\mathbb{I} - KH)A(k-1)}_{\hat{A}(k)} \hat{x}_m(k-1)$$

$$+ \underbrace{(\mathbb{I} - KH)}_{\hat{B}(k)} u(k-1) + K\bar{z}(k)$$

KF not guaranteed to be optimal – non-linear estimators may do better

4.1.1 Asymptotic Properties of KF

Model assume **Tl System, stationary Distributions** (A, H, Q, R constants)

Change of Variance

$$P_p(k+1) = \left(AP_p(k)A^\top \right) + Q$$

$$- \left(AP_p(k)H^\top \right) \left(HP_p(k)H^\top + R \right)^{-1} \left(HP_p(k)A^\top \right)$$

$$\textbf{Scalar Case} \; P_p(k+1) = \frac{a^2 r P_p(k)}{h^2 P_p(k) + r} + q =: f(P_p(k))$$

Summary *Variance Converges to unique steady-state sol'n provided either $|a| < 1$ or if $|a| \geq 1 \Rightarrow h \neq 0$ and $q > 0$*

If slope for small P_∞ is $> 1 \Rightarrow$ unstable ($|a| > 1$)

4.1.2 Detectability / Stabilizability

Detectability Conditions

Pair (A, H) detectable

$$\Leftrightarrow \text{ For deterministic LTI Sys. } (x(k) = Ax(k-1), z(k) = Hx(k)) \\ \lim_{k \rightarrow \infty} z(k) = 0 \Rightarrow \lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$$

$$\Leftrightarrow \begin{bmatrix} A & -\lambda \mathbb{I} \\ H & \end{bmatrix} \text{ is full rank for all } \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1 \text{ (PBH-Test)}$$

$$\Leftrightarrow \text{Eigenvalues of } A - LH \text{ (or } (\mathbb{I} - LH)A) \text{ can be placed within unit circle by suitable choice of } L \in \mathbb{R}^{n \times m}$$

Stabilizability Conditions

Pair (A, B) stabilizable

$$\Leftrightarrow \text{ For deterministic LTI Sys. } (x(k) = Ax(k-1) + Bu(k-1)) \\ \exists u(0:k-1) \text{ s.t. } \lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$$

$$\Leftrightarrow [A - \lambda \mathbb{I} \; B] \text{ full rank } \forall \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1 \text{ (PBH-Test)}$$

$$\Leftrightarrow \text{Eigenvalues of } A - BK \text{ (or } (\mathbb{I} - BK)A) \text{ can be placed within unit circle by suitable choice of } K \in \mathbb{R}^{m \times n}$$

NOTE Stabilizability is dual of detectability, (A, B) stabilizable

Observability / Reachability

if $(A^\top, B^\top)$ detectable

Observability Matrix

implies detectability

Reachability Matrix

implies stabilizability

$$\mathcal{O} = \begin{bmatrix} H \\ HA^{n-1} \end{bmatrix}$$

$$\mathcal{R} = \begin{bmatrix} A & A^{n-1}B \end{bmatrix}$$

4.1.3 Steady-State KF

DARE – Discrete Algebraic Ricatti Equation (P_∞ is P_p)

$$P_\infty = AP_\infty A^\top + Q - AP_\infty H^\top (HP_\infty H^\top + R)^{-1} HP_\infty A^\top$$

$$\textbf{Steady-State KF Gain} \qquad K_\infty = P_\infty H^\top (HP_\infty H^\top + R)^{-1}$$

Filter Equations Linear Time-Invariant System

$$\hat{x}(k) = \underbrace{(\mathbb{I} - K_\infty H)A}_{\hat{A}} \hat{x}(k-1) + \underbrace{(\mathbb{I} - K_\infty H)}_{\hat{B}} u(k-1) + K_\infty \bar{z}(k)$$

Error dynamics (also holds for non ss-KF) $e(k) = x(k) - x_f(k)$

$$e(k) = (\mathbb{I} - K_\infty H)Ae(k-1) + (\mathbb{I} - K_\infty H)v(k-1) - K_\infty w(k)$$

$$\mathbb{E}\{e(k)\} = (\mathbb{I} - K_\infty H)A \; \mathbb{E}\{e(k-1)\}$$

- $(\mathbb{I} - K_\infty H)A$ must be stable (all $|\lambda| < 1$) for e not to diverge

- $\mathbb{E}\{e(k)\} \xrightarrow{k \rightarrow \infty} 0$ iff $(\mathbb{I} - K_\infty H)A$ stable

Theorem for $R > 0, Q \geq 0, G$ s.t. $Q = GG^\top$

$$\Leftrightarrow (A, H) \text{ is detectable, } (A, G) \text{ is stabilizable}$$

$$\Leftrightarrow \text{DARE has unique pos. semidef. sol. } P_\infty \geq 0, \text{ resulting } (\mathbb{I} - K_\infty H)A \text{ is stable and } \lim_{k \rightarrow \infty} P_p(k) = P_\infty \quad \forall P_p(1) \geq 0$$

(A,H) Detectable: can observe all unstable modes

(A,G) Stabilizable: noise excites unstable modes

Note if $Q > 0 \Rightarrow (A, G)$ always stabilizable If $Q = 0$ multiple valid solutions can be possible

4.2 Extended Kalman Filter

Extension of KF to non-linear systems:

$$x(k) = q_{k-1}(x(k-1), v(k-1))$$
$$z(k) = h_k(x(k), w(k))$$

$x(0), v(\cdot), w(\cdot)$ mutually ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

EKF Equations

Initialization: $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step

$$\hat{x}_p(k) = q_{k-1}(\hat{x}_m(k-1), 0)$$

$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) \\ + L(k-1)Q(k-1)L^\top(k-1)$$

$$A(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial x}, \quad L(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial v}$$

S2 – A Posteriori Update / Measurement Update (at k)

$$K = P_p H^\top \left(H P_p H^\top + M R M^\top \right)^{-1}$$

$$\hat{x}_m = \hat{x}_p + K(\bar{z} - h_k(\hat{x}_p(k), 0))$$

$$P_m = (\mathbb{I} - K(k)H(k)) P_p(k)$$

$$H(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial x}, \quad M(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial w}$$

Intuition

Process Update – predict the mean state estimate fwd using NL process model and update the var. according to lin. eqns

Measurement Update – correct for mismatch between $\bar{z}(k)$ and prediction $h_k(\hat{x}_p(k), 0)$, and correct var. according to lin. eqn

Caution

- Process & measurement noise **assumed** 0-mean
- A, L, H, M lin. about current $\hat{x}_m \rightsquigarrow$ **cannot** compute offline
- Prior update only accurate if $\mathbb{E}\{\cdot\}$ & $q_{k-1}(\cdot)$ commute
- EKF Variables DO NOT capture true conditional mean & var.**

4.2.1 Hybrid EKF

System Process – non-lin, CT, **Measurement** – non-lin, DT

$$\dot{x}(k) = q(x(t), v(t), t) \qquad x[k] := x(kT)$$

$$z[k] = h_k(x[k], w[k]) \qquad \mathbb{E}\{w[k]\} = 0, \text{Var}\{w[k]\} = R$$

CT White Noise $\delta(t)$ = Dirac pulse

$$\mathbb{E}\{v(t)\} = 0 \text{ and } \mathbb{E}\{v(t)v^\top(t+\tau)\} = \mathcal{Q}_c\delta(\tau)$$

True CT White noise does not exist $\rightarrow$ Would require infinite power

Hybrid EKF Equations

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$

S1 – Prior Update / Prediction Step (Solve ODEs)

Solve, for $(k-1)T \leq t \leq kT$ and $\hat{x}((k-1)T) = \hat{x}_m[k-1]$

$$\dot{\hat{x}}(t) = q(\hat{x}(t), 0, t) \quad \Rightarrow \hat{x}_p[k] := \hat{x}(kT)$$

Solve, for $(k-1)T \leq t \leq kT$ and $P((k-1)T) = P_m[k-1]$

$$\dot{P}(t) = AP + PA^\top + LQ_cL^\top \quad \Rightarrow P_p[k] := P(kT)$$

with

$$A(t) := \frac{\partial q}{\partial x}(\hat{x}(t), 0, t), \qquad L(t) := \frac{\partial q}{\partial v}(\hat{x}(t), 0, t)$$

S2 – A Posteriori Update / Measurement Update

Measurement model still DT – S2 identical to standard EKF

Caution

- Process & measurement noise **assumed** 0-mean
- ODEs typically solved numerically: Accuracy $\uparrow$ = Comp. Cost $\uparrow$

5 Particle Filter

5.1 Monte Carlo Sampling

Samples N DRVs y^n of samples of $y \in \mathcal{Y} = \{1, \dots, \bar{Y}\} \sim p_y$

Helper Variables

$$s_i^n = \delta(i - y^n) = \begin{cases} 1 & \text{if } y^n = i \\ 0 & \text{otherwise} \end{cases} \quad \begin{matrix} i = 1, \dots, \bar{Y} \\ n = 1, \dots, N \end{matrix}$$

Average – s_i converges to $p_y(i)$ by Law of Large Numbers

$$\frac{1}{N} \sum_{n=1}^N s_i^n \xrightarrow{N \rightarrow \infty} \mathbb{E}\{s_i^n\} = p_y(i) \qquad p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \bar{s}_i^n$$

Approximation

For $y \in \mathcal{Y}$, it holds $p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \delta(i - \bar{y}^n), \quad i \in \mathcal{Y}$

For $x = g(y)$, with $x \in \mathcal{X} := g(\mathcal{Y})$, it holds

$$p_x(j) \approx \frac{1}{N} \sum_{n=1}^N \delta(j - g(\bar{y}^n)), \quad j \in g(\mathcal{Y})$$

Monte Carlo Sampling CRV

Helper Variables

$$s_a^n = \int_a^{a+\Delta y} \delta(i - y^n) d\xi = \begin{cases} 1 & \text{if } a \leq y^n < a + \Delta y \\ 0 & \text{otherwise} \end{cases}$$

Average – s_a converges to $p_y(\xi)$ by Law of Large Numbers

$$\int_a^{a+\Delta y} \frac{1}{N} \sum_{n=1}^N \delta(\xi - y^n) d\xi \xrightarrow{N \rightarrow \infty} \int_a^{a+\Delta y} p_y(\xi) d\xi$$

Approximation

$$\text{For } y \in \mathcal{Y}, \text{ it holds } p_y(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{y}^n) \forall \xi$$

$$\text{For } x = g(y), \text{ it holds } p_x(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - g(\bar{y}^n)) \forall \xi$$

5.2 Modeling

System *Non-linear, discrete-time*:

$$x(k) = q_{k-1}(x(k-1), v(k-1))$$

$$z(k) = h_k(x(k), w(k))$$

$x(0), \{v\}, \{w\}$ mutually indep. can be D- or CRVs, known PDFs

Objective

Bayesian State Estimator with Monte Carlo Sampling

Init:	$x_m(0)$	$:= x(0)$
S1:	$x_p(k)$	$:= q_{k-1}(x_m(k-1), v(k-1))$
S2:	$z_m(k)$	$:= h_k(x_p(k), w(k))$
	$p_{x_m(k)}(\xi)$	$:= p_{x_p(k) z_m(k)}(\xi \bar{z}(k)) \quad \forall \xi$

Similar proof for $p_{x_p(k)}(\xi), p_{x_m(k)}(\xi) \forall \xi$ as for KF aux. var.

5.3 Particle Filter - Equations

Initialization Draw N samples $\{\bar{x}_m^n(0)\}$ from $p_{x(0)}$

S1 – Prior Update / Prediction Step

Apply process equation to the particles $\{\bar{x}_m^n(k-1)\}$

$$\bar{x}_p^n(k) := q_{k-1}(\bar{x}_m^n(k-1), \bar{v}^n(k-1)), \quad n = 1, \dots, N$$

Requires N noise samples from $p_{v(k-1)}$

S2 – A Posteriori Update / Measurement Update

Scale each particle by measurement likelihood:

$$\beta_n = \alpha p_{z(k)|x(k)}(\bar{z}(k)|\bar{x}_p^n(k)), \qquad n = 1, \dots, N$$

with α normalization s.t $\sum_{n=1}^N \beta_n = 1$

$$\alpha = \left(\sum_{n=1}^N p_{z_m(k)|x_p(k)}(\bar{z}(k)|\bar{x}_p^n(k)) \right)^{-1}$$

Resample to get N posterior particles $\{\bar{x}_m^n(k)\}$, w/ eq. weights

Repeat N times:
- Select random number r uniformly on $(0, 1)$ - Pick particle $\bar{n}$ s.t $\sum_{n=1}^{\bar{n}-1} \beta_n < r \leq \sum_{n=1}^{\bar{n}} \beta_n$

New particles all have equal weight

$$p_{x_m(k)}(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{x}_m^n(k)), \quad \forall \xi$$

Intuition

Process Update – Propagate N particles through system dynamics. Provided N large, $\{x_m^n(k-1)\} \approx p_{x_m(k-1)} \rightsquigarrow \{\bar{x}_p^n(k)\} \approx p_{x_p(k)}$

Measurement Update – treat each particle separately. At points of high prior, we have many particles. Posterior has same particles, but scaled by measurement likelihood.

5.4 Roughening

Sample Impoverishment *Lumpiness* – when we resample, only retain subset of particles. For finite N this is problem.

Roughening

Perturb particles **after** resampling

$$\bar{x}_m^n(k) \leftarrow \bar{x}_m^n(k) + \Delta x^n(k)$$

$\Delta x^n(k)$ drawn from 0-mean, finite-var. distribution

Example Method for Δx^n : σ_i stand. dev. of $\Delta x_i^n(k)$

$$\sigma_i = K E_i N^{-\frac{1}{d}}$$

K: tuning parameter, typically $K \ll 1$

d: dimension of state space

E_i : max inter-sample variability $\max_{n_1, n_2} |\bar{x}_{m,i}^{n_1} - \bar{x}_{m,i}^{n_2}|$

$N^{-\frac{1}{d}}$: node spacing on grid

6 Observer Based Control

6.1 LTI Observer

LTI System

$$x(k) = Ax(k-1) + Bu(k-1) + v(k-1)$$

$$z(k) = Hx(k) + w(k)$$

v, w 0-mean noise CRVs

$$\mathbb{E}\{\hat{x}(k)\} \xrightarrow{k \rightarrow \infty} \mathbb{E}\{x(k)\}, \text{Var}\{\hat{x}(k)\} \text{ bounded}$$

Luenberger Observer

Same structure as steady-state KF:

$$\hat{x}(k) = A\hat{x}(k-1) + Bu(k-1) + K(\bar{z}(k) - \hat{z}(k)) \\ = (\mathbb{I} - K_{\infty}H)A\hat{x}(k-1) \\ + (\mathbb{I} - K_{\infty}H)Bu(k-1) + K_{\infty}\bar{z}(k) \\ \hat{z}(k) = H(A\hat{x}(k-1) + Bu(k-1))$$

Error $e(k) = x(k) - \hat{x}(k)$

in absence of noise $\bar{z}(k) = z(k)$

$$e(k) = (\mathbb{I} - KH)Ae(k-1)$$

- $(\mathbb{I} - KH)A$ must be stable (all $|\lambda| < 1$) for $e \xrightarrow{k \rightarrow \infty} 0$
- Such K exists iff (A, H, A) detectable $\Leftrightarrow (A, H)$ detectable

Observer Design

If (A, H) detectable, construct K

Pole Placement: place λ of e -dynamics of observer modes at desired locations

Steady State KF: Yields optimal K (min steady-state MSE) given statistics of v, w . Also need (A, G) stabilizable, $\text{Var}\{v(k)\} = GG^\top$

Alternative Formulation

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + K(\bar{z}(k) - \hat{z}(k)) \\ \hat{z}(k) = H\hat{x}(k) \rightarrow \text{delayed measurement}$$

Error Dynamics

$$e(k+1) = (A - KH)e(k)$$

6.2 Static State Feedback Control

System Model

LTI, no noise, deterministic, perf. state info $z(k) = x(k)$:

Feedback Law	CL Dynamics
$u(k) = Fx(k) = Fz(k)$	$x(k) = (A + BF)x(k-1)$

- $(A + BF)$ must be stable (all $|\lambda| < 1$)
- Such F exists $\Leftrightarrow (A, B)$ stabilizable

Controller Design

Pole Placement: place (controllable) poles at des. CL loc.

LQR: find F that min. quad. cost $(\bar{Q} = \bar{Q}^\top \geq 0, \bar{R} = \bar{R}^\top \geq 0)$

$$J_{\text{LQR}} = \sum_{k=0}^{\infty} x^\top(k)\bar{Q}x(k) + u^\top(k)\bar{R}u(k)$$

LQR Design

Assuming (A, B) stabil., (A, G) detect.

$$F = -(B^\top PB + \bar{R})^{-1}B^\top PA$$

DARE

Same as KF but $A \rightarrow A^\top, H \rightarrow B^\top, Q \rightarrow \bar{Q}, R \rightarrow \bar{R}$

$$P = A^\top PA + \bar{Q} - A^\top PB(B^\top PB + \bar{R})^{-1}B^\top PA$$

6.3 Separation Principle

Stable Observer + Controller $\rightsquigarrow u(k) = F\hat{x}(k)$

Dynamics

$$\begin{bmatrix} x(k) \\ e(k) \end{bmatrix} = \begin{bmatrix} A + BF & -BF \\ 0 & (\mathbb{I} - KH)A \end{bmatrix} \begin{bmatrix} x(k-1) \\ e(k-1) \end{bmatrix}$$

Separation Principle

- $(\mathbb{I} - KH)A$ and $(A + BF)$ stable $\rightsquigarrow$ **Overall system stable**

- Still stable with noise, Generalizable to time-varying case
- Does **NOT** hold for nonlinear systems

Separation Theorem

Optimal strategy for control problem

$$J_{\text{LQG}} = \lim_{N \rightarrow \infty} \mathbb{E} \left\{ \frac{1}{N} \sum_{k=0}^{N-1} \left[x^\top(k) \bar{Q} x(k) + u^\top(k) \bar{R} u(k) \right] \right\}$$

1. Design steady state KF to provide $\hat{x}(k)$ **(no dependency on $\bar{Q}, \bar{R}$)**
2. Design optimal state-FB law $u(k) = Fx(k)$ for determ. LQR problem
minimizes J_{LQR} w/o noise **(no dependency on noise statistics Q, R)**
3. Put together